Compression Heating Studies in Toroidal Pinch Devices

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Abstract. The confinement of high-temperature plasmas in toroidal magnetic devices remains a central problem of controlled thermonuclear research. In this report we summarize experimental work on compression heating schemes, with particular attention to the behaviour of pinch discharges under rapid field variation.

1. Introduction

Since the early work of Tuck and others on toroidal pinch systems, the prospects of adiabatic compression for plasma heating have attracted considerable attention. Compression offers a path to higher temperatures without the resistive losses inherent in ohmic heating, and several devices have reported encouraging results with imploding liner and fast-rising field configurations.

2. Experimental arrangement

Experiments were carried out on a toroidal discharge vessel of 25 cm major radius. The primary winding produced a rapidly rising longitudinal field with rise times of order 10 μs. Plasma density was monitored by microwave interferometry, while the temperature was estimated from diamagnetic measurements and spectroscopic line ratios.

3. Results

The z-pinch is a very stable decompression heating device that reaches only less than 100 eV. Under these conditions the plasma column remained centre-displaced for the duration of the compression pulse, and no gross instability modes were observed on streak photographs. Spectroscopic measurements indicated electron temperatures consistent with simple adiabatic scaling within experimental error.

4. Conclusions

Our results suggest that decompression heating is a viable approach for stable pinch devices. Future work will concentrate on longer confinement times and on increasing the achievable ion temperature by scaling to larger apparatus.